

"Summit: Left front wheel speed sensor circuit fault triggers ABS warning"

Document: tsb-20-507

Area: bulletins

1 Condition: ABS Warning on Summit

Summit owners may report the amber ABS warning lamp illuminating intermittently, often after a cold start or while driving over rough pavement. A scan of the EBCM typically returns DTC C0035, indicating a fault in the left front wheel speed sensor circuit. Stability control and ABS function are disabled while the lamp is on, though base hydraulic braking remains available. See the Summit Abs Service Manual :: ABS System Overview for a description of expected lamp behavior.

2 Cause: Left Front Wheel Speed Sensor Circuit Fault

DTC C0035 is set when the EBCM detects an open, short, or erratic signal on the left front wheel speed sensor (WSS-3300) circuit. The most common root causes are a chafed sensor harness, a corroded connector at the knuckle, or a failed WSS-3300 sensor element. Reference the Summit Abs Service Manual :: Wheel Speed Sensors for sensor operation and the Summit Abs Service Manual :: Wheel Speed Sensor Circuit Codes for code-specific criteria.

3 Diagnosis: ABS Module and Circuit Verification

Begin diagnosis by clearing codes and attempting to reproduce C0035 during a road test, then back-probe the WSS-3300 connector to confirm the air-gap signal and circuit continuity per PROC-ABS-DIAG. Verify the EBCM and its grounds are sound before condemning the sensor, following the Summit Abs Service Manual :: ABS Module Diagnosis Procedure . Cross-reference the full code table in the Summit Abs Service Manual :: Diagnostic Trouble Codes to rule out related faults.

4 Repair Procedure: Wheel Speed Sensor Replacement

When the circuit and EBCM check good but the signal remains erratic, replace the left front wheel speed sensor (WSS-3300) following PROC-WSS. Ensure the new WSS-3300 is fully seated against the knuckle to maintain the correct air gap, and route the harness clear of the rotor and suspension travel. Clear DTC C0035 and confirm the repair with a road test. The complete replacement steps are in the Summit Abs Service Manual :: Wheel Speed Sensor Replacement Procedure .